

ASYMPTOTIC EXPANSIONS FOR DISTRIBUTIONS OF CHARACTERISTIC ROOTS OF COVARIANCE MATRICES*

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SUMMARY

An alternative approach which appears simpler is given for obtaining the distribution of characteristic roots of S and $S_1 S_2^{-1}$ where $S \sim W_p(n^{-1}\Sigma, n)$, $S_i \sim W_p(\Sigma_i, n_i)$, $i = 1, 2$, and S_1 and S_2 are independently distributed. This approach also helps to establish the order of each term which so far has not been established in earlier papers on this subject.

1. Introduction

An asymptotic expansion for the distribution of the characteristic roots of a sample covariance matrix $S \sim W_p(n^{-1}\Sigma, n)$ which arises in principal component analysis, among others, was given by G.A. Anderson (1965) up to $O(n^{-2})$ when all the roots of Σ are distinct. James (1969) extended the result to the case when the last $(p-q)$ roots of Σ are equal and gave an expansion up to $O(n^{-1})$ while Chattopadhyay and Pillai (1973) gave an expansion up to $O(n^{-2})$. The method utilized by these authors is to integrate over the orthogonal matrix H around the neighbourhood of the point at which the maximum of the density is obtained, and then they claimed that each term is of the given order *without* actually establishing it. Even, some alternative methods developed recently by Sugiura (1976) and the partial differential equation method developed by Constantine and Muirhead (1976) have not succeeded in removing this difficulty. The results in this paper are achieved by using the transformation $H = -I + 2(I - \frac{1}{2}Q)^{-1}$, rather than the transformation $H = \exp(Q)$, where Q is a skew-symmetric matrix. This gives a simple expres-

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sion for the Jacobian of the transformation. Also the integration over Q is easier here which helps in establishing the order of each term.

The above method is extended to obtain the asymptotic distribution of the roots of $S_1 S_2^{-1}$ where $S_i \sim W(\Sigma_i, n_i)$ and S_1 and S_2 are independently distributed and $n = n_1 + n_2$ is large. This problem has been considered by Chang (1970), Chattopadhyay and Pillai (1973), Chattopadhyay, Pillai and Li (1976), Sugiura (1976) and Constantine and Muirhead (1976) but they did not establish the order of each term. While the extension of the techniques of this paper to other problems in multivariate analysis are under investigation, Carter and Srivastava (1977) have established the results for the roots of non-central Wishart matrix, a result not available in the literature so far.

2. Preliminary Results

Let S be a $p \times p$ positive definite random symmetric matrix. Then there exists an orthogonal matrix H with non-negative diagonal elements (H unique) such that $S = H D_\ell H'$ where $D_\ell = \text{diag}(\ell_1, \dots, \ell_p)$ and $\ell_1 > \ell_2 > \dots > \ell_p > 0$ with probability one. The following results are available in the literature (see, for example Srivastava and Khatri, 1977). Here $J(X \rightarrow Y)$ denotes the Jacobian of the transformation from X to Y .

Lemma 2.1

Let $H' = (H'_1, H'_2)$ be a $p \times p$ orthogonal matrix with positive diagonal elements and $H_1 : q \times p, q \leq p$. Let

$$\alpha_p(D_\ell) = \prod_{i=1}^{p-1} \prod_{j=i+1}^p (\ell_i - \ell_j)$$

$$f_{p,p}(H) = J(H'(dH) \rightarrow dH)$$

$$f_{q,p}(H_1) = J((dH_1)H' \rightarrow dH_1) \text{ and}$$

$$\Gamma_p\left(\frac{1}{2}a\right) = \left\{ \prod_{i=1}^p \Gamma\left(\frac{a-i+1}{2}\right) \right\} \pi^{\frac{1}{4}p(p-1)}.$$

Then

$$(2.1) \quad J(S \rightarrow D_\ell, H) = \alpha_p(D_\ell) f_{p,p}(H),$$

$$(2.2) \quad \int_{H'H=I} f_{p,p}(H) dH = \pi^{\frac{1}{2}p^2} / \Gamma_p\left(\frac{1}{2}p\right),$$

and

$$(2.3) \int_{H_1 H_1' = I_q} f_{q,p}(H_1) dH_1 = \pi^{\frac{1}{2}pq} / \Gamma_q(\frac{1}{2}p), \quad q \leq p.$$

Corollary 2.1

Let $D = \{H_2 \mid \text{given } H_1 : H_2' H_2 = I - H_1' H_1\}$ where $H' = (H_1', H_2')$ is a random orthogonal matrix. Then

$$(2.4) \int_D f_{p,p}(H) dH = f_{q,p}(H_1) \pi^{\frac{1}{2}(p-q)^2} / \Gamma_{p-q}(\frac{1}{2}(p-q)).$$

Proof

Since

$$f_{p,p}(H) = J((dH)H' \rightarrow dH)$$

$$= J((dH_1)H_1' \rightarrow dH_1) J((dH_2)H_2' \rightarrow dH_2 \mid H_1),$$

$$\pi^{-\frac{1}{2}p^2} \Gamma_p(\frac{1}{2}p) f_{p,p}(H) \text{ is the joint pdf of } H_1 \text{ and } H_2 \text{ (see (2.2)) and } \pi^{-\frac{1}{2}pq} \Gamma_q(\frac{1}{2}p) f_{q,p}(H_1)$$

is the marginal pdf of H_1 . Hence, the result follows from the fact that

$$\Gamma_p(\frac{1}{2}p) = \pi^{\frac{1}{2}q(p-q)} \Gamma_q(\frac{1}{2}p) \Gamma_{p-q}(\frac{1}{2}(p-q)).$$

Corollary 2.2

If

$$H = -I + 2(I - \frac{1}{2}T)^{-1}$$

where T is a $p \times p$ skew-symmetric matrix and H is an orthogonal matrix, then

$$(2.5) J(H \rightarrow T) = |I - \frac{1}{2}T|^{-(p-1)} / f_{p,p}(H).$$

Proof

Note that

$$J(H \rightarrow T) = J(dH \rightarrow dT)$$

$$= J(dH \rightarrow H'dH) J(H'dH \rightarrow dT).$$

Since

$$H'dH = ((I - \frac{1}{2}T)^{-1})' (dT) ((I - \frac{1}{2}T)^{-1}),$$

we get

$$J(H'dH \rightarrow dT) = |I - \frac{1}{2} T|^{-(p-1)}.$$

The result now follows since $f_{p,p}(H) = J(H'dH \rightarrow dH)$.

Next, we consider the case when the number of independent elements in H is not $\frac{1}{2} p(p-1)$ but $pq - \frac{1}{2} q(q+1)$.

Let $H = \begin{pmatrix} H_1 \\ H_2 \end{pmatrix} : p \times p$ be an orthogonal matrix such that H_2 is functionally dependent on $H_1 : q \times p$ where $H_1 H_1' = I_q$ and $H_2 H_2' = I_{p-q}$. Then the number of independent elements in H is the same as in H_1 which is $pq - \frac{1}{2} q(q+1)$. And although $R = (dH)H'$ is still a skew-symmetric matrix, the number of independent elements is now only $pq - \frac{1}{2} q(q+1)$, the same as in $(dH_1)H_1'$. Let Q_{11} be a $q \times q$ skew symmetric matrix and Q_{12} be a $q \times (p-q)$ matrix. Then the number of independent elements in Q_{11} is $\frac{1}{2} q(q-1)$ and the number of elements in Q_{12} is $q(p-q)$. Thus, the total number of independent elements in Q_{11} and Q_{12} together is $\frac{1}{2} q(2p-q-1) = pq - \frac{1}{2} q(q+1)$. Hence, we can write the orthogonal matrix H as

$$(2.6) \quad H = -I + 2(I - \frac{1}{2} Q)^{-1},$$

where

$$(2.7) \quad Q = \begin{pmatrix} Q_{11} & Q_{12} \\ -Q_{12}' & 0 \end{pmatrix}.$$

Similarly, since $(dH_1)H_1' = ((dH_1)H_1', (dH_1)H_2')$ where $(dH_1)H_1'$ is a skew symmetric matrix, we can write

$$(2.8) \quad (dH_1)H_1' = ((dH_1)H_1', (dH_1)H_2') = (T_{11}, T_{12})$$

where $T_{11} : q \times q$ is a skew symmetric matrix and T_{12} is a $q \times (p-q)$ matrix. Hence

$$(2.9) \quad (T_{11}, T_{12}) = (dH_1)H_1' = A(I_q, \frac{1}{2} Q_{12})(dQ)(I + \frac{1}{2} Q)^{-1},$$

where

$$(2.10) \quad A = (I_q - \frac{1}{2} Q_{11} + \frac{1}{4} Q_{12} Q_{12}')^{-1}.$$

Thus

$$(2.11) \quad T_{11} = A(dQ_{11}) A' - A Q_{12} (dQ_{12})' A' + A (dQ_{12}) Q'_{12} A'$$

and

$$(2.12) \quad T_{12} = A(dQ_{12}) - T_{11} Q'_{12}.$$

Hence

$$\begin{aligned} J(H_1 \rightarrow Q_{11}, Q_{12}) &= J(dH_1 \rightarrow T_{11}, T_{12}) J(T_{11}, T_{12} \rightarrow dQ_{11}, dQ_{12}) \\ &= (1/f_{q,p}(H_1)) |A|_+^{(p-1)}. \end{aligned}$$

Hence, we get the following

Lemma 2.2

Let $H_1 : q \times p$ be a random matrix such that $q \leq p$ and $H_1 H'_1 = I_q$. Let

$$(2.13) \quad H_1 = (-I_q + 2(I_q - \frac{1}{2} Q_{11} + \frac{1}{4} Q_{12} Q'_{12})^{-1}, \\ (I - \frac{1}{2} Q_{11} + \frac{1}{4} Q_{12} Q'_{12})^{-1} Q_{12})$$

that is H is given by (2.6). Then

$$(2.14) \quad J(H_1 \rightarrow Q_{11}, Q_{12}) = [1/f_{q,p}(H_1)] |A|_+^{(p-1)},$$

where A is given in (2.10), and $|A|_+$ denotes the positive value of the determinant of A .

Lemma 2.3

Let $T = (t_{ij})$ be a $p \times p$ skew-symmetric matrix with pdf $f(T)$ given by

$$f(T) = \prod_{i=1}^{p-1} \prod_{j=i+1}^p \{ (2\pi h_{ij})^{-\frac{1}{2}} \exp(-\frac{1}{2} t_{ij}^2 / h_{ij}) \},$$

that is t_{ij} 's $i = 1, 2, \dots, p-1, j = i+1, \dots, p$ are independent normally distributed as $N(0, h_{ij})$, $h_{ij} > 0$. Let $\{\sqrt{-1} \lambda_\alpha(T), -\sqrt{-1} \lambda_\alpha(T)\}$, $\alpha = 1, 2, \dots, r$, $r \leq [\frac{1}{2} p]$, where $[a]$ denotes the largest integer contained in $[0, a]$, be the non-zero characteristic roots of T and let $\lambda_{\max}^*(T) = \max_\alpha |\lambda_\alpha(T)|$ and $D = \{T: \lambda_{\max}^*(T) > (4n)^{\frac{1}{2}}\}$. If $g(T)$ is a function of T such that

$$|g(T)| \leq g_0 (\lambda_{\max}^*(T))^s$$

where g_0 is a constant and $s \geq 0$, then

$$|\int_D g(T) f(T) dT| = O((cn)^{\frac{1}{2}s + \frac{1}{4}p(p-1)-1} e^{-cn})$$

where $c = (\max_{ij} h_{ij})^{-1}$.

Proof

Let $\text{ch}_{\max}(A)$ denote the maximum ch. root of A . Then, since

$$|\lambda_{\alpha}(T)|^2 \leq \text{ch}_{\max}(T'T) \leq \text{tr } T'T = 2 \sum_{i=1}^{p-1} \sum_{j=i+1}^p (t_{ij}^2/h_{ij}) h_{ij},$$

it follows that

$$(\lambda_{\max}^+(T))^2 < 2c^{-1} \sum_{i=1}^{p-1} \sum_{j=i+1}^p (t_{ij}^2/h_{ij}) \equiv 2c^{-1} y \equiv 4c^{-1} x,$$

and $D \subset D_0$ where $D_0 = \{T: x > cn\}$. From the pdf of T , it follows that $y \sim x_1^2$. Hence $x \sim \frac{1}{2} x_1^2$ and $\frac{1}{2} p(p-1)$.

$$\begin{aligned} |\int_D g(T) f(T) dT| &\leq g_0 \int_D [\lambda_{\max}^+(T)]^s f(T) dT \\ &\leq g_0 \{\Gamma(\frac{1}{4} p(p-1))\}^{-1} (4/c)^{\frac{1}{2}s} \int_{cn}^{\infty} x^{\frac{1}{2}s + \frac{1}{4}p(p-1)} e^{-x} dx \end{aligned}$$

or

$$\begin{aligned} |\int_D g(T) f(T) dT| &\leq g_0 \{\Gamma(p(p-1)/4)\}^{-1} (4/c)^{\frac{1}{2}s} (cn)^{\frac{1}{2}s + \frac{1}{4}p(p-1)} \\ &\quad \int_1^{\infty} x^{\frac{1}{2}s + \frac{1}{4}p(p-1)} \exp(-cnx) dx. \end{aligned}$$

Observe that if $x \geq 1$,

$$x^{\frac{1}{2}s + \frac{1}{4}p(p-1)-1} \leq x^{[\frac{1}{2}s + \frac{1}{4}p(p-1)]}$$

where $[\frac{1}{2}s + \frac{1}{4}p(p-1)] =$ greatest integer contained in $[0, \frac{1}{2}s + \frac{1}{4}p(p-1)]$. Hence

$$\frac{x^{\frac{1}{2}s + \frac{1}{4}p(p-1)-1}}{\Gamma([\frac{1}{2}s + \frac{1}{4}p(p-1)] + 1)} \leq \frac{x^{[\frac{1}{2}s + \frac{1}{4}p(p-1)]}}{\{[\frac{1}{2}s + \frac{1}{4}p(p-1)]\}!} < e^x,$$

and this gives

$$| \int_D g(T) f(T) dT | < \frac{g_0 \Gamma([\frac{1}{2}s + \frac{1}{4}p(p-1)] + 1) (4/c)^{\frac{1}{2}s} (cn)^{\frac{1}{2}s + \frac{1}{4}p(p-1)}}{\Gamma(p(p-1)/4) (cn-1)} \cdot e^{-cn}$$

which proves Lemma 2.3.

Note 1

If some of the h_{ij} 's are zero, then the corresponding t_{ij} 's = 0 and they do not occur in Lemma 2.3. Hence on the right side of the result, $\frac{1}{2}p(p-1)$ should be reduced by the number of zero h_{ij} 's or t_{ij} 's.

Lemma 2.4

Let A and B be $p \times p$ positive definite matrices and T be a $p \times p$ skew-symmetric matrix. Then for non-negative integers i and j,

$$|\text{tr}(AT^i BT^j)| \leq p^2 (\text{ch}_{\max} A) (\text{ch}_{\max} B) (\lambda_{\max}^+(T))^{i+j}$$

where $\text{ch}_{\max}(\cdot) = \text{maximum characteristic root of } (\cdot)$ and $\lambda_{\max}^+(T) = \max |\text{ch}(T)|$.

Proof

Since T is a skew-symmetric matrix, there exists a unitary matrix U such that

$$T = \sqrt{(-1)} U D_w U^*$$

where U^* is the complex conjugate of U, $D_w = \text{diag}(w_1, w_2, \dots, w_p)$, w_α 's are real and $\sqrt{(-1)}w_\alpha$, $\alpha = 1, 2, \dots, p$ are the eigen values of T.

Let $A_1 = U^*AU$ and $B_1 = U^*BU$. Then

$$\begin{aligned} \text{tr}(AT^i BT^j) &= (-1)^{(i+j)/2} \text{tr}(A U D_w^i U^* B U D_w^j U^*) \\ &= (-1)^{(i+j)/2} \text{tr}(A_1 D_w^i B_1 D_w^j) \\ &= (-1)^{(i+j)/2} \sum_{k, \ell} a_{k\ell} w_\ell^i b_{\ell k} w_k^j \end{aligned}$$

where $A_1 = (a_{k\ell})$ and $B_1 = (b_{\ell k})$ are $p \times p$ Hermitian p.d. Since $\lambda_{\max}(T) = \max_\ell |w_\ell|$, we have

$$|\text{tr}(AT^iBT^j)| \leq \sum_{k,\ell} |a_{k\ell}| |b_{\ell k}| |w_\ell|^i |w_k|^j \\ \leq (\lambda_{\max}^+(T))^{i+j} \sum_{k,\ell} |a_{k\ell}| |b_{\ell k}|.$$

Since A_1 and B_1 are Hermitian p.d. matrices, we get

$$|a_{k\ell}|^2 \leq a_{kk} a_{\ell\ell}, |b_{k\ell}|^2 \leq b_{kk} b_{\ell\ell}$$

and if $\text{ch}_{\max}(\cdot)$ = maximum characteristic root of $(\cdot)$,

$$a_{kk} \leq (\text{ch}_{\max} A), b_{kk} \leq \text{ch}_{\max} B.$$

Using the above results, we get

$$|\text{tr}(AT^iBT^j)| \leq p^2 (\text{ch}_{\max} A) (\text{ch}_{\max} B) (\lambda_{\max}^+(T))^{i+j},$$

which proves Lemma 2.4.

3. Asymptotic Distribution of the Roots of S

Let $S \sim W_p(n^{-1}D_\lambda, n)$ where $D_\lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ and $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p > 0$. In this section we derive the asymptotic distribution of the roots $\ell_1, \ell_2, \dots, \ell_p$ of S . Subsection 3.2 deals with the case when the population roots λ_i 's are distinct and subsection 3.2 considers the case when some of the roots are equal.

3.1 Population Roots Distinct

Let

$$(3.1) \quad S = HD_\ell H' \text{ and } H = -I + 2[I - (4n)^{-\frac{1}{2}} T]^{-1}$$

where H is an orthogonal matrix, T is a skew-symmetric matrix and $D = \text{diag}(\ell_1, \dots, \ell_p)$ with $\ell_1 > \ell_2 > \dots > \ell_p$, the roots of S . From (2.1) and (2.5), the Jacobian of the transformation

$$(3.2) \quad J(S \rightarrow D_\ell, T) = n^{-\frac{1}{4} p(p-1)} \alpha_p(D_\ell) |I - (4n)^{-\frac{1}{2}} T|^{-(p-1)}.$$

Let

$$(3.3) \quad c_{ij} = (\lambda_j^{-1} - \lambda_i^{-1})(\ell_i - \ell_j), T = (t_{ij}) \text{ with } t_{ii} = 0, t_{ij} = -t_{ji}.$$

$$(3.4) \quad f_1(T) = \prod_{i=1}^{p-1} \prod_{j=i+1}^p \{ (2\pi)^{-\frac{1}{2}} c_{ij}^{\frac{1}{2}} (\exp - \frac{1}{2} c_{ij} t_{ij}^2) \},$$

$$(3.5) \quad f_2(D_\ell | D_\lambda) = \left\{ \left(\frac{1}{2} n \right)^{\frac{1}{2} p n - \frac{1}{4} p(p-1)} / \prod_{j=1}^p \Gamma(\frac{n-j+1}{2}) [\alpha_p(D_\ell) / \alpha_p(D_\lambda)]^{\frac{1}{2}} \right. \\ \left. |D_\lambda^{-1} D_\ell|^{\frac{1}{2} (n-p-1)} |D_\lambda^{-1}| \exp(-\frac{1}{2} n \operatorname{tr} D_\lambda^{-1} D_\ell) \right\}$$

and

$$(3.6) \quad f_3(T, D_\ell | D_\lambda) = |I - (4n)^{-\frac{1}{2}} T|^{-(p-1)} \exp P,$$

where

$$(3.7) \quad P = \frac{1}{2} n \operatorname{tr} D_\lambda^{-1} D_\ell - \frac{1}{2} n \operatorname{tr} D_\lambda^{-1} H D_\ell H' + \frac{1}{2} \sum_{i=1}^{p-1} \sum_{j=i+1}^p c_{ij} t_{ij}^2 \\ = \frac{1}{2} n \operatorname{tr} D_\lambda^{-1} D_\ell - \frac{1}{2} n \operatorname{tr} D_\lambda^{-1} H D_\ell H' + \frac{1}{2} (\operatorname{tr} D_\lambda^{-1} D_\ell T^2 - \operatorname{tr} D_\lambda^{-1} T D_\ell T),$$

and where H is given by (3.1). Since the pdf of S is given by

$$(3.8) \quad \left\{ \left(\frac{1}{2} n \right)^{\frac{1}{2} p n} / \Gamma_p(\frac{1}{2} n) \right\} |D_\lambda|^{-\frac{1}{2} n} |S|^{\frac{1}{2} (n-p-1)} \exp - \frac{1}{2} n D_\lambda^{-1} S,$$

the joint pdf of D_ℓ and T is given by

$$(3.9) \quad f_1(T) f_2(D_\ell | D_\lambda) f_3(T, D_\ell | D_\lambda).$$

By Lemma 2.3,

$$(3.10) \quad P(\lambda_{\max}^+(T) \geq (4n)^{\frac{1}{2}}) = \int_D f_1(T) dT = O(e^{-cn} (cn)^{\frac{1}{4} p(p-1)-1})$$

where $\lambda_{\max}^+(T)$ denotes the maximum value of $|\operatorname{ch}(T)|$, $\operatorname{ch} T$ being an eigen value of T, $D = \{T: \lambda_{\max}^+(T) \geq (4n)^{\frac{1}{2}}\}$ and $c = (\min_{i,j} c_{ij})$. Hence, we can expand $f_3(T, D_\ell | D_\lambda)$ by using

$$(3.11) \quad H = I + \sum_{j=0}^{\infty} \left(\frac{1}{2} \right)^j n^{-\frac{1}{2} (j+1)} T^{j+1} \\ = I + E \text{ (say),}$$

and

$$(3.12) \quad \|I - (4n)^{-\frac{1}{2}} T\|^{-(p-1)} = \exp \left(\sum_{j=1}^{\infty} (p-1) 2^{-2j-1} n^{-j} \operatorname{tr} T^{2j/j} \right).$$

Thus, for large n (since $\operatorname{tr} D_{\lambda}^{-1} D_{\ell} T^{2i+1} = 0$)

$$(3.13) \quad P = -\frac{1}{2} n \operatorname{tr} D_{\lambda}^{-1} E D_{\ell} E' - n \operatorname{tr} D_{\lambda}^{-1} D_{\ell} \left(\sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^{2j+1} n^{-j-1} T^{2j+2} \right) \\ + \frac{1}{2} (\operatorname{tr} D_{\lambda}^{-1} D_{\ell} T^2 - \operatorname{tr} D_{\lambda}^{-1} T D_{\ell} T) \\ = \sum_{j=1}^{\infty} n^{-\frac{1}{2}j} \{T\}_{j+2}$$

where

$$\{T\}_{j+2} = \left(\frac{1}{2}\right)^{2m+1} \sum_{\alpha=0}^m (-1)^{\alpha} \operatorname{tr} D_{\lambda}^{-1} T^{2m-\alpha+2} D_{\ell} T^{\alpha+1} \quad \text{if } j = 2m+1 \\ = - \left(\frac{1}{2}\right)^{2m+1} [\operatorname{tr} D_{\lambda}^{-1} D_{\ell} T^{2m+2} - (-1)^m \operatorname{tr} D_{\lambda}^{-1} T^{m+1} D_{\ell} T^{m+1}] \\ + \left(\frac{1}{2}\right)^{2m} \sum_{\alpha=0}^{m-1} (-1)^{\alpha} \operatorname{tr} D_{\lambda}^{-1} T^{2m-\alpha+1} D_{\ell} T^{\alpha+1} \quad \text{if } j = 2m.$$

Thus from (3.12) and (3.13), we can write

$$(3.15) \quad f_3(T, D_{\ell} | D_{\lambda}) = 1 + \sum_{j=1}^{\infty} n^{-\frac{1}{2}j} g_j(T)$$

where

$$g_j(T) = \sum_{\alpha=1}^j \sum_{(j_1, \dots, j_{\alpha}: \sum_{t=1}^{\alpha} j_t = j)} (\alpha!)^{-1} \{h_{j_1+2}(T) h_{j_2+2}(T) \dots h_{j_{\alpha}+2}(T)\} \\ \text{with } j_t \geq 1$$

and

$$h_j(T) = \{T\}_{2m+3} \quad \text{if } j = 2m+3, \\ = \{T\}_{2m+2} + m^{-1}(p-1)\left(\frac{1}{2}\right)^{2m+1} \operatorname{tr} T^{2m} \quad \text{if } j = 2m+2.$$

By Lemma 2.4, $|\{T\}_j| \leq (\frac{1}{2})^{j-2} \{[\frac{1}{2}j] + 1\} p^2 (\ell_1/\lambda_p) (\lambda_{\max}^*(T))^j$, for $j = 2, 3, \dots$.

Hence, it is easy to see that for $j = 1, 2, \dots$, $|g_j(T)| \leq g_{j0} \{\lambda_{\max}(T)\}^{3j}$ where g_{j0} depends on j , $(\ell_1/\lambda_p)p^2$ and $(p-1)$, and $\lambda_{\max}(T) > 1$.

Thus $g_{2m+1}(T)$ is a polynomial function of odd degree in T so that

$$g_{2m+1}(-T) = -g_{2m+1}(T). \text{ Hence}$$

$$E g_{2m+1}(T) = 0,$$

where E stands for the expectation over T with pdf given by $f_1(T)$ in (3.4). Hence, by Lemma 2.3,

$$\begin{aligned} (3.16) \quad E f_3(T, D_\ell | D_\lambda) &= 1 + \sum_{j=1}^{\infty} n^{-j} E g_{2j}(T) \\ &= 1 + \sum_{j=1}^{\infty} n^{-j} a_j \text{ (say).} \end{aligned}$$

We shall obtain only a_1 explicitly.

We have

$$(3.17) \quad g_2(T) = h_4(T) + \frac{1}{2} \{h_3(T)\}^2 = \{T\}_4 + \frac{1}{2} (\{T_3\})^2 + \frac{p-1}{8} \text{tr } T^2$$

where

$$\begin{aligned} 2 \{T_3\} &= \text{tr } D_\lambda^{-1} T^2 D_\ell T \\ &= \sum_{i < j < k} [-\lambda_i^{-1} (\ell_j - \ell_k) + \lambda_j^{-1} (\ell_i - \ell_k) - \lambda_k^{-1} (\ell_i - \ell_j)] t_{ij} t_{jk} t_{ki}, \\ 8 \{T\}_4 &= -\text{tr } D_\lambda^{-1} D_\ell T^4 - \text{tr } D_\lambda^{-1} T^2 D_\ell T^2 + 2 \text{tr } D_\lambda^{-1} T^3 D_\ell T \\ &= \sum_{\substack{i \neq k \\ j \neq \alpha}} [\lambda_i^{-1} (\ell_i - 2\ell_j + \ell_k) t_{ij} t_{jk} t_{\alpha k} t_{i\alpha} \end{aligned}$$

$$\begin{aligned}
& - \sum_{i \neq j} [2\lambda_i^{-1}(\ell_i - \ell_j)] t_{ij}^4 \\
& - \sum_{i \neq j \neq k} [\lambda_i^{-1}(\ell_i - 2\ell_j + \ell_k) + 2\lambda^{-1}(\ell_j - \ell_i)] t_{ij}^2 t_{jk}^2,
\end{aligned}$$

and

$$\text{tr } T^2 = -2 \sum_{i < j} t_{ij}^2.$$

Hence

$$\begin{aligned}
8E \{T\}_4 &= 6 \sum_{i < j} (c_{ij})^{-1} + \sum_{i < j < k} \{(c_{ij}c_{jk})^{-1}(2c_{ij} + 2c_{jk} - c_{ik}) \\
&+ (c_{ij}c_{ik})^{-1}(2c_{ij} + 2c_{ik} - c_{jk}) + (c_{ik}c_{jk})^{-1}(2c_{ik} + 2c_{jk} - c_{ij})\} \\
&= (4p-2) \sum_{i < j} (c_{ij})^{-1} - \sum_{i < j < k} (c_{ij}c_{jk}c_{ik})^{-1} (c_{ij}^2 + c_{ik}^2 + c_{jk}^2), \\
4E \{T\}_3^2 &= \sum_{i < j < k} (c_{ij}c_{jk}c_{ki})^{-1} [-\lambda_i^{-1}(\ell_j - \ell_k) + \lambda_j^{-1}(\ell_i - \ell_k) - \lambda_k^{-1}(\ell_i - \ell_j)]^2 \\
&= \sum_{i < j < k} (c_{ij}c_{jk}c_{ik})^{-1} [(c_{ij}^2 + c_{ik}^2 + c_{jk}^2) - 2(c_{ij}c_{ik} + c_{ij}c_{jk} + c_{ik}c_{jk})]
\end{aligned}$$

and

$$E(\text{tr } T^2) = -2 \sum_{i < j} (c_{ij})^{-1}.$$

Hence

$$\begin{aligned}
a_1 &= E[g_2(T)] = \frac{1}{8} E[-\text{tr } D_\lambda^{-1} D_\ell T^4 + \text{tr } D_\lambda^{-1} T^2 D_\ell T^2 + 2 \text{tr } D_\lambda^{-1} T^3 D_\ell T \\
&+ (\text{tr } D_\lambda^{-1} T^2 D_\ell T)^2 + (p-1) \text{tr } T^2] \\
&= \frac{1}{2} \sum_{i < j} (c_{ij})^{-1}.
\end{aligned}$$

Thus, we have established the following

Theorem 3.1

Let $S \sim W_p(n^{-1} D_\lambda, n)$, $D_\lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 > \lambda_2 > \dots > \lambda_p$ and let $\ell_1 > \ell_2 > \dots > \ell_p$ be the characteristic roots of S . If $c_{ij} = (\lambda_i - \lambda_j)(\ell_i - \ell_j)/\lambda_i \lambda_j$, $i \neq j$ and $f_2(D_\ell | D_\lambda)$ is defined by (3.5), then for large value of n , the density of $\ell_1, \ell_2, \dots, \ell_p$ is given by

$$f_2(D_\ell | D_\lambda) [1 + (2n)^{-1} \sum_{i < j} (c_{ij})^{-1} + O(n^{-2})].$$

3.2 Last (p-q) Population Roots Equal

In this section we will consider the case when the last $(p-q)$ roots of D_λ are equal to λ (say), that is, when

$$(3.18) \quad D_\lambda = \text{diag}(\Lambda, \lambda I_{p-q}) \text{ where } \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_q)$$

and $\lambda_1 > \lambda_2 > \dots > \lambda_q$. This problem has been considered by James (1969). Here, we give an alternative method and establish the order of each term as well.

Let us now write the orthogonal matrix H as $H = \begin{pmatrix} H_1 \\ H_2 \end{pmatrix}$ where $H_1 : q \times p$ and $H_2 : (p-q) \times p$. This gives

$$(3.19) \quad \text{tr } D_\lambda^{-1} H D_\ell H' = \text{tr } H' D_\lambda^{-1} H D_\ell = \lambda^{-1} \text{tr } D_\ell + \text{tr } \Lambda_1 H_1 D_\ell H_1',$$

where

$$(3.20) \quad \Lambda_1 = (\Lambda^{-1} - \lambda^{-1} I_q).$$

Thus from (3.8), we get the joint pdf of D_ℓ and H as

$$\begin{aligned} & [(\frac{1}{2}n)^{\frac{1}{2}pn} / \Gamma_p(\frac{1}{2}n)] |D_\lambda|^{-\frac{1}{2}n} |D_\ell|^{\frac{1}{2}(n-p-1)} \alpha_p(D_\ell) f_{p,p}(H) \\ & \exp - \frac{n}{2} (\lambda^{-1} D_\ell + \Lambda_1 H_1 D_\ell H_1'). \end{aligned}$$

Integrating out H_2 with the help of (2.4) and making the transformation

$$\begin{aligned} H_1 &= [I_q + 2(I_q - (4n)^{-\frac{1}{2}} T_{11} + (4n)^{-1} T_{12} T_{12}')^{-1}, \\ & \quad n^{-\frac{1}{2}} (I_q - (4n)^{-\frac{1}{2}} T_{11} + (4n)^{-1} T_{12} T_{12}')^{-1} T_{12}], \end{aligned}$$

we get the joint pdf of T_{11} , T_{12} and D_ℓ with the help of Lemma 2.2 as,

$$(3.21) \quad \frac{n^{\frac{1}{2}} p(n-q) + \frac{1}{4} q(q+1) \frac{1}{n^{\frac{1}{2}} (p-q)^2}}{2^{\frac{1}{2}} p^n \Gamma_{p-q}(\frac{p-q}{2}) \Gamma_p(\frac{1}{2} n) |D_\lambda|^{n/2}} \\ \cdot \frac{|D_\ell|^{\frac{1}{2}(n-p-1)} \alpha_p(D_\ell) \text{etr} - \frac{n}{2} (\lambda^{-1} D_\ell + \Lambda_1 H_1 D_\ell H_1')}{\| \lambda_q - \frac{1}{2} \sqrt{n} T_{11} + \frac{1}{4n} T_{12} T_{12}' \|^{(p-1)}}$$

Let

$$(3.22) \quad c_{ij} = (\lambda_i^{-1} - \lambda_j^{-1})(\ell_i - \ell_j) \quad i \neq j, \text{ to } (i, j) = 1, 2, \dots, q \\ = (\lambda^{-1} - \lambda_i^{-1})(\ell_i - \ell_j) \text{ for } j = q+1, \dots, p \text{ and } i = 1, 2, \dots, q;$$

$$(3.23) \quad f_{11}(T_{11}, T_{12}) = \left(\prod_{i=1}^{q-1} \prod_{j=i+1}^q 2\pi c_{ij} \right)^{\frac{1}{2}} \left(\prod_{j=q+1}^p \prod_{i=1}^q 2\pi c_{ij} \right)^{\frac{1}{2}} \\ \exp \left[-\frac{1}{2} \sum_{i \neq j}^q c_{ij} t_{ij}^2 - \frac{1}{2} \sum_{j=q+1}^p \sum_{i=1}^q c_{ij} t_{ij}^2 \right],$$

$$(3.24) \quad f_{01}(D_\ell | D_\lambda) = \\ \frac{\left(\frac{n}{2}\right)^{\frac{1}{2} p(n-1) + \frac{1}{4} q(q+1)} \frac{1}{n^{\frac{1}{2}} (p-q)} \alpha_p(D_\ell) |D_\ell|^{\frac{1}{2}(n-p-1)} \text{etr} - \frac{1}{2} n D_\lambda^{-1} D_\ell}{\left\{ \prod_{j=1}^p \Gamma(\frac{n-j+1}{2}) \right\} \left\{ \prod_{j=1}^{p-1} \Gamma(\frac{p-q-j+1}{2}) \right\} |D_\lambda|^{\frac{1}{2}(n-p+1)} \lambda^{\frac{1}{2}(p-q)(p-q-1)}} \\ \cdot \left\{ \prod_{i < j=1}^q (\lambda_i - \lambda_j)(\ell_i - \ell_j) \right\}^{-\frac{1}{2}} \left\{ \prod_{j=q+1}^p \prod_{i=1}^q (\lambda_i - \lambda)(\ell_i - \ell_j) \right\}^{-\frac{1}{2}}$$

and

$$(3.25) \quad g_1(T_{11}, T_{12}, D_\ell | D_\lambda) = f_3(T, D_\ell | D_\lambda)$$

with

$$(3.26) \quad T = \begin{pmatrix} T_{11} & T_{12} \\ -T'_{12} & 0 \end{pmatrix}, \quad D_\lambda = \begin{pmatrix} \Lambda & 0 \\ 0 & \lambda I_{p-q} \end{pmatrix} \quad \text{and}$$

$f_3(T, D_\ell | D_\lambda)$ defined in (3.7). Note that

$$(3.27) \quad \sum_{i < j} c_{ij} t_{ij}^2 + \sum_{j=q+1}^p \sum_{i=1}^q c_{ij} t_{ij}^2 \\ = \text{tr } \Lambda_1 T_{11} D_{\ell 1} T_{11} - \text{tr } \Lambda_1 D_{\ell 1} T_{11}^2 + \text{tr } \Lambda_1 (D_{\ell 1} T_{12} T'_{12} - T_{12} D_{\ell 2} T'_{12}),$$

where

$$D_\ell = \text{diag}(D_{\ell 1}, D_{\ell 2}), T_{11} = (t_{ij}) \text{ with } t_{ij} = -t_{ji} \text{ for } i \neq j,$$

$$i, j = 1, 2, \dots, q \text{ and } t_{ii} = 0, i = 1, 2, \dots, q \text{ and}$$

$$T_{12} = (t_{ij+q}), i = 1, 2, \dots, q \text{ and } j = 1, 2, \dots, p-q.$$

Then the joint pdf of T_{11} , T_{12} and D_ℓ is given by

$$(3.28) \quad f_1(T_{11}, T_{12}) f_{01}(D_\ell | D_\lambda) g_1(T_{11}, T_{12}, D_\ell | D_\lambda).$$

From Note 1 at the end of Lemma 2.3, we get

$$P(\lambda_{\max}^+(T) > (4n)^{\frac{1}{2}}) = O(e^{-cn_{(cn)}^{\frac{1}{2}} p q - \frac{1}{4} q(q+1) - 1})$$

where T is now defined in (3.26), $c = \min_{i,j} (c_{ij})$ and $\lambda_{\max}^+(T) = \max |ch(T)|$.

Hence, $g_1(T_{11}, T_{12}, D_\ell | D_\lambda)$ can be expanded by using

$$H_1 = [I_q + 2 \sum_{i=1}^{\infty} ((4n)^{-\frac{1}{2}} T_{11} - (4n)^{-1} T_{12} T'_{12})^i, \\ n^{-\frac{1}{2}} (I + \sum_{i=1}^{\infty} ((4n)^{-\frac{1}{2}} T_{11} - (4n)^{-1} T_{12} T'_{12})^i) T_{12}],$$

and a similar expansion of $|I_q - (4n)^{-\frac{1}{2}} T_{11} + (4n)^{-1} T_{12} T'_{12}|^{-(p-1)}$. Thus, as in the distinct roots case, it can be shown with the help of Lemmas 2.3 and 2.4 that

$$\begin{aligned} \text{Eg}_1(T_{11}, T_{12}, D_q | D_\lambda) &= 1 + \sum_{j=1}^{\infty} n^{-j} \text{Eg}_{2j}(T) \\ &= 1 + \sum_{j=1}^{\infty} n^{-j} a_j \text{ (say)} \end{aligned}$$

where g_{2j} is the same as given in (3.15) with T defined in (3.26). Thus a_j 's can be obtained by putting $t_{ij} = 0$ for $i, j = q+1, \dots, p$. We shall calculate only a_1 . We have

$$\begin{aligned} 4E\{T_3\}^2 &= \left(\sum_{k=q+1}^p \sum_{i < j=1}^q + \sum_{i < j < k=1}^q \right) (c_{ij} c_{jk} c_{ki})^{-1} \\ &\quad [(c_{ij}^2 + c_{ik}^2 + c_{jk}^2) - 2(c_{ij} c_{ik} + c_{ij} c_{jk} + c_{ik} c_{jk})], \\ E(\text{tr } T^2) &= -2 \sum_{i < j=1}^q (c_{ij})^{-1} - 2 \sum_{j=q+1}^p \sum_{i=1}^q (c_{ij})^{-1}, \\ 8E\{T\}_4 &= 6 \left(\sum_{i < j=1}^q (c_{ij})^{-1} + \sum_{j=q+1}^p \sum_{i=1}^q (c_{ij})^{-1} \right) \\ &\quad + \left(\sum_{k=q+1}^p \sum_{i < j=1}^q + \sum_{i < j < k=1}^q \right) (c_{ij} c_{jk})^{-1} (2c_{ij} + 2c_{jk} - c_{ki}) \\ &\quad + (c_{ij} c_{ik})^{-1} (2c_{ij} + 2c_{ik} - c_{jk}) + (c_{ik} c_{jk})^{-1} (2c_{ik} + 2c_{jk} - c_{ij}) \} \\ &\quad + 2(p-q-1) \sum_{j=q+1}^p \sum_{i=1}^q (c_{ij})^{-1} \\ &= (4p-2) \sum_{i < j=1}^q c_{ij}^{-1} + (2p+2q) \sum_{j=q+1}^p \sum_{i=1}^q (c_{ij})^{-1} \\ &\quad - \left(\sum_{k=q+1}^p \sum_{i < j=1}^q + \sum_{i < j < k=1}^q \right) (c_{ij}^2 + c_{ik}^2 + c_{jk}^2) / (c_{ij} c_{ik} c_{jk}). \end{aligned}$$

Hence,

$$a_1 = \text{Eg}_2(T) = \frac{1}{2} \sum_{i < j=1}^q c_{ij}^{-1} + \frac{1}{2} \sum_{j=q+1}^p \sum_{i=1}^q (c_{ij})^{-1}.$$

Thus, we get the following

Theorem 3.2

Let $S \sim W_p(n^{-1}D_\lambda, n)$, $D_\lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ $\lambda_1 > \dots > \lambda_q > \lambda_{q+1} = \dots = \lambda_p = \lambda$ (say), and let $\ell_1 > \ell_2 > \dots > \ell_p$ be the characteristic roots of S . If $c_{ij} = (\lambda_i - \lambda_j)(\ell_i - \ell_j) / \lambda_i \lambda_j$ for $i < j = 1, 2, \dots, q$ and $c_{ij} = (\lambda_i - \lambda)(\ell_i - \ell_j) / \lambda \lambda_i$ for $i = 1, 2, \dots, q$ and $j = q+1, \dots, p$ and $f_{01}(D_\ell | D_\lambda)$ is defined by (3.24), then for large value of n , the density of $\ell_1, \dots, \ell_p$ is defined by

$$f_{01}(D_\ell | D_\lambda) [1 + \frac{1}{2n} (\sum_{i < j=1}^q (c_{ij})^{-1} + \sum_{j=q+1}^p \sum_{i=1}^q (c_{ij})^{-1}) + O(n^{-2})].$$

Clearly from Theorem 3.1 and Theorem 3.2, we can extend the result to multiple roots, i.e., where $D_\lambda = (\delta_1 I_{p_1}, \delta_2 I_{p_2}, \dots, \delta_k I_{p_k})$, $\sum_{i=1}^k p_i = p$ and at least one of $p_1, p_2, \dots, p_k$ is 1.

Remark 3.1

It should be noted that the constant term in Theorem 3.2 is different from the one given by James (1969).

4. Asymptotic Distribution of the Roots of $S_1 S_2^{-1}$

Let S_1 and S_2 be independently distributed as $W_p(\Sigma_i, n_i)$, $i = 1, 2$. Let

$$\ell_1 > \ell_2 > \dots > \ell_p > 0$$

be the characteristic roots of $S_1 S_2^{-1}$ and $\lambda_p \geq \lambda_{p-1} \geq \dots \geq \lambda_1 > 0$ be the characteristic roots of $\Sigma_2 \Sigma_1^{-1}$. In this section we derive the asymptotic distribution of the roots $\ell_1, \dots, \ell_p$.

4.1 Distinct Population Roots Case

Here we consider the case when the population roots are distinct, that is,

$$\lambda_p > \lambda_{p-1} > \dots > \lambda_1 > 0.$$

The joint pdf of $\ell_1, \ell_2, \dots, \ell_p$ can be written as

$$(4.1) \quad n^{-\frac{1}{4}p(p-1)} (\beta_p(\frac{1}{2}n_1, \frac{1}{2}n_2))^{-1} |D_\ell|^{\frac{1}{2}(n_1-p-1)} |D_\lambda|^{\frac{1}{2}n_1} \alpha_p(D_\ell)$$

$$f|I + D_\lambda H D_\ell H'|^{-\frac{1}{2}n} |I - (4n)^{-\frac{1}{2}} T|^{-(p-1)dT}$$

where

$$H = -I + 2(I - (4n)^{-\frac{1}{2}} T)^{-1}, \quad T \text{ is skew-symmetric,}$$

$$n = n_1 + n_2, \beta_p(\frac{1}{2}n_1, \frac{1}{2}n_2) = \Gamma_p(\frac{1}{2}n_1) \Gamma_p(\frac{1}{2}n_2) / \Gamma_p(\frac{1}{2}n).$$

Let

$$(4.2) \quad D_1 = (D_\lambda^{-1} + D_\ell)^{-1} \text{ and } A_1(T) = D_1^{\frac{1}{2}} T D_\ell D_1^{\frac{1}{2}} + D_1^{\frac{1}{2}} D_\ell T' D_1^{\frac{1}{2}}.$$

Then, using

$$H = I + 2(4n)^{-\frac{1}{2}} T + 2(4n)^{-1} T^2 A \text{ with } A = (I - 2(4n)^{-\frac{1}{2}} T)^{-1},$$

one can write

$$(4.3) \quad |I + D_\lambda H D_\ell H'|^{-\frac{1}{2}n} |I + D_\lambda D_\ell|^{\frac{1}{2}n} \\ = |I + 2(4n)^{-\frac{1}{2}} A_1(T) + 2(4n)^{-1} D_1^{\frac{1}{2}} (T^2 A D_\ell + D_\ell A' T^2 + 2T A D_\ell A' T') D_1^{\frac{1}{2}}|^{-\frac{1}{2}n}$$

Notice that $A_1(T)$ is a symmetric matrix with zero diagonal elements and (4.3) gives

$$(4.4) \quad \lim_{n \rightarrow \infty} (|I + D_\lambda H D_\ell H'|^{-\frac{1}{2}n} |I + D_\lambda D_\ell|^{\frac{1}{2}n}) = \exp(-\frac{1}{2} \{T\}_2)$$

where $\{T\}_2 = \text{tr } A_2(T) - \frac{1}{2} \text{tr } \{A_1(T)\}^2$ with

$$(4.5) \quad A_2(T) = \frac{1}{2} D_1^{\frac{1}{2}} (T^2 D_\ell + D_\ell T^2) D_1^{\frac{1}{2}} + D_1^{\frac{1}{2}} T D_\ell T' D_1^{\frac{1}{2}}.$$

Using (4.2) and (4.5), we get

$$(4.6) \{T\}_2 = -\text{tr } D_1 D_\ell T T' + \text{tr } D_1 T D_\ell T' + \text{tr } D_1 D_\ell T D_1 D_\ell T' - \text{tr } D_1 T D_\ell^2 D_1 T' \\ = \sum_{i=1}^{p-1} \sum_{j=i+1}^p c_{ij} t_{ij}^2.$$

with

$$(4.7) c_{ij} = (\ell_i - \ell_j)(\lambda_j - \lambda_i)/(1 + \ell_i \lambda_i)(1 + \ell_j \lambda_j) \text{ for } i < j, i, j = 1, 2, \dots, p.$$

Hence, we can rewrite (4.1) as

$$(4.8) f_0(D_\ell | D_\lambda) g(D_\ell)$$

where

$$(4.9) f_0(D_\ell | D_\lambda) = (2\pi/n)^{\frac{1}{4}p(p-1)} \{\alpha_p(D_\ell)/\alpha_p(D_\lambda)\}^{\frac{1}{2}} \{\beta_p(\frac{1}{2}n_1, \frac{1}{2}n_2)\}^{-1} \\ |D_\lambda|^{\frac{1}{2}n_1} |I + D_\lambda D_\ell|^{-\frac{1}{2}(n-p+1)} |D_\ell|^{\frac{1}{2}(n_1-p-1)}$$

and

$$(4.10) g(D_\ell) = \int f_1(T) f_4(T, D_\ell | D_\lambda) dT$$

with $f_1(T)$ being defined by (3.4) in which c_{ij} 's are defined by (4.7) and

$$(4.11) f_4(T, D_\ell | D_\lambda) = \exp \left(\frac{1}{2} \{T\}_2 \right) |I + D_\ell D_\lambda|^{\frac{1}{2}n} |I + D_\lambda H D_\ell H'|^{-\frac{1}{2}n} \\ \cdot |I - (4n)^{-\frac{1}{2}} T|^{-(p-1)}$$

Observe that (4.11) can be expanded provided $\lambda_{\max}^+(T) = \max |ch(T)| < (4qn)^{\frac{1}{2}}$ where $q (> 0)$ is a function of λ_i 's and ℓ_j 's. With respect to the density $f_1(T)$, we have by Lemma 2.3,

$$P(\lambda_{\max}^+(T) \geq (4qn)^{\frac{1}{2}}) = O(e^{-qcn} (qcn)^{\frac{1}{4}p(p-1)-1})$$

where $c = \min_{i,j} c_{ij}$. Hence, using (4.3), (4.11) can be expanded and it can be written as

$$(4.12) f_4(T, D_\ell | D_\lambda) = \exp \left(\sum_{j=1}^{\infty} n^{-\frac{1}{2}j} \{T\}_{j+2} \right) |I - (4n)^{-\frac{1}{2}} T|^{-(p-1)}$$

where for $j = 3, 4, \dots$

$$(4.13) \quad \{T\}_j = \sum_{\alpha=1}^j \frac{(-1)^\alpha}{2^\alpha} \sum_{j_1, \dots, j_\alpha} \text{tr} \{A_{j_1}(T) A_{j_2}(T) \dots A_{j_\alpha}(T)\}$$

with $j_t \geq 1$ and $\sum_{t=1}^{\infty} j_t = j$

with $A_1(T)$ being defined in (4.2) and for $j = 1, 2, \dots$

$$(4.14) \quad A_{j+1}(T) = \left(\frac{1}{2}\right)^j D_1^{\frac{1}{2}} (T^{j+1} D_\ell + D_\ell (T^{j+1})') D_1^{\frac{1}{2}} \\ + \left(\frac{1}{2}\right)^{j-1} \sum_{\alpha=0}^{j-1} D_1^{\frac{1}{2}} T^j D_\ell (T^{j-\alpha})' D_1^{\frac{1}{2}}.$$

Notice that when j is odd

$$\{T\}_j = -\{-T\}_j.$$

Hence, as in section 3, we can finally obtain the joint density of $\ell_1, \ell_2, \dots, \ell_p$ when n is large as

$$(4.15) \quad f_0(D_\ell | D_\lambda) [1 + \sum_{j=1}^{\infty} n^{-1} a_j]$$

where $f_0(D_\ell | D_\lambda)$ is defined in (4.9), and $a_j = E g_{2j}(T)$ with $g_j(T)$ as defined in (3.15). The expectation is over the distribution of t_{ij} 's distributed independently $N(0, 1/c_{ij})$. We shall obtain the explicit expression for a_1 only. Notice that

$$(4.16) \quad 2 \{T\}_3 = -\text{tr} A_3(T) + \text{tr} A_2(T) A_1(T) - \frac{1}{3} \text{tr} (A_1(T))^3 \\ = \text{tr} D_1 T^2 D_\ell T + \text{tr} D_1 T D_1 D_\ell^2 T^2 + 2 \text{tr} D_1 D_\ell T D_1 T D_\ell T \\ + \text{tr} D_1 T D_1 D_\ell T D_1 D_\ell^2 T$$

and

$$(4.17) \quad 2 \{T\}_4 = -\text{tr} A_4(T) + \text{tr} A_3(T) A_1(T) + \frac{1}{2} \text{tr} (A_2(T))^2 \\ - \text{tr} A_2(T) (A_1(T))^2 + \frac{1}{4} \text{tr} (A_1(T))^2 \\ = -\frac{1}{4} (\text{tr} D_1 D_\ell T^4 - 2 \text{tr} D_1 T D_\ell T^3 + \text{tr} D_1 T^2 D_\ell T^2) \\ + \frac{1}{2} (\text{tr} D_1 D_\ell T D_1 D_\ell T^3 - \text{tr} D_1 T D_1 D_\ell^2 T^3)$$

$$\begin{aligned}
& -(\text{tr } D_1 D_\ell \text{TD}_1 \text{TD}_\ell T^2 - \text{tr } D_1 \text{TD}_1 D_\ell \text{TD}_\ell T^2) \\
& + \frac{1}{4} (\text{tr } D_1 D_\ell T^2 D_1 D_\ell T^2 + \text{tr } D_1 T^2 D_1 D_\ell^2 T^2 \\
& + 2 \text{tr } D_1 \text{TD}_\ell \text{TD}_1 \text{TD}_\ell T - 4 \text{tr } D_1 D_\ell \text{TD}_\ell \text{TD}_1 T^2) \\
& - (\text{tr } D_1 D_\ell \text{TD}_1 D_\ell \text{TD}_1 D_\ell T^2 + \text{tr } D_1 \text{TD}_1 D_\ell \text{TD}_1 D_\ell^2 T^2 \\
& - \text{tr } D_1 \text{TD}_1^2 D_\ell \text{TD}_1 D_\ell^2 T - \text{tr } D_1 D_\ell \text{TD}_1 \text{TD}_1 D_\ell^2 T^2 \\
& + \text{tr } D_1 \text{TD}_1 D_\ell^2 \text{TD}_1 \text{TD}_\ell T + \text{tr } D_1 D_\ell \text{TD}_1 \text{TD}_1 D_\ell \text{TD}_\ell T \\
& - 2 \text{tr } D_1 \text{TD}_1 D_\ell \text{TD}_1 D_\ell \text{TD}_\ell T) \\
& + \frac{1}{2} (\text{tr } D_1 D_\ell \text{TD}_1 D_\ell \text{TD}_1 D_\ell \text{TD}_1 D_\ell T + \text{tr } D_1 \text{TD}_1 D_\ell^2 \text{TD}_1 \text{TD}_1 D_\ell^2 T \\
& - 2 \text{tr } D_1 \text{TD}_1 D_\ell \text{TD}_1 D_\ell \text{TD}_1 D_\ell^2 T + 2 \text{tr } D_1 \text{TD}_1 D_\ell \text{TD}_1 D_\ell^2 \text{TD}_1 D_\ell T \\
& - 2 \text{tr } D_1 \text{TD}_1 D_\ell^2 \text{TD}_1 D_\ell \text{TD}_1 D_\ell T) \\
& = -\frac{1}{4} \sum_{i,j,k,\alpha} f(i,j,k,\alpha) t_{ij} t_{jk} t_{k\alpha} t_{\alpha i}
\end{aligned}$$

with

$$\begin{aligned}
f(i,j,\alpha) = & (w_i \ell_i - 2w_i \ell_j + w_i \ell_k) - 2(w_i \ell_i w_j \ell_j - w_i w_j \ell_j^2) \\
& + 4(w_i \ell_i w_j \ell_k - w_i w_j \ell_j \ell_k) - (w_i \ell_i w_k \ell_k + w_i w_k \ell_k^2 \\
& + 2w_i w_k \ell_j \ell_\alpha - 4w_i \ell_i \ell_j w_k) + 4(w_i \ell_i w_j \ell_j w_k \ell_k \\
& + w_i w_j \ell_j w_k \ell_k^2 - w_i w_j \ell_j w_k \ell_k - w_i \ell_i w_j w_k \ell_k^2 + w_i w_j \ell_j^2 w_k \ell_\alpha \\
& + w_i w_j \ell_i \ell_k w_k \ell_\alpha - 2w_i w_j \ell_j w_k \ell_k \ell_\alpha) - 2(w_i \ell_i w_j \ell_j w_k \ell_k w_\alpha \ell_\alpha \\
& + w_i \ell_j^2 w_j w_k w_\alpha \ell_\alpha^2 - 2w_i w_j \ell_j w_k \ell_k w_\alpha \ell_\alpha^2 + 2w_i w_j \ell_j w_k \ell_k^2 w_\alpha \ell_\alpha \\
& - 2w_i w_j \ell_j^2 w_k \ell_k w_\alpha \ell_\alpha),
\end{aligned}$$

and $w_i = \lambda_i / (1 + \lambda_i \ell_i)$, $i = 1, 2, \dots, p$. It can be shown after some heavy algebra that

$$4E\{T\}_3^2 = \sum_{i < j < k} \frac{1}{c_{ij}c_{jk}c_{ik}} [c_{ij}^2 + c_{ik}^2 + c_{jk}^2 - 2(c_{ij}c_{ik} + c_{ij}c_{jk} + c_{ik}c_{jk}) - 4c_{ij}c_{ik}c_{jk}]$$

and

$$E\{T\}_4 = \frac{3}{4} \sum_{i < j} c_{ij}^{-2} (c_{ij} + c_{ij}^2) + \frac{1}{8} \sum_{i < j < k} \left[\frac{1}{c_{ij}c_{jk}} (2c_{ij} + 2c_{jk} - c_{ik} + 4c_{ij}c_{jk}) + (c_{ij}c_{ik})^{-1} (2c_{ij} + 2c_{ik} - c_{jk} + 4c_{ij}c_{ik}) + (c_{ik}c_{jk})^{-1} (2c_{ik} + 2c_{jk} - c_{ij} + 4c_{ik}c_{jk}) \right].$$

Hence,

$$a_1 = \frac{1}{2} E\{T\}_3^2 + E\{T\}_4 = \frac{1}{2} \sum_{i < j} (c_{ij})^{-1} + \frac{1}{24} p(p-1)(4p+1).$$

Thus, we have the following

Theorem 4.1

Let S_i ($i = 1, 2$) be independently distributed as $W_p(\Sigma_i, n_i)$, ($i = 1, 2$) and let the ch. roots of $S_1 S_2^{-1}$ and $\Sigma_2 \Sigma_1^{-1}$ be $D_\ell = \text{diag}(\ell_1, \dots, \ell_p)$, $\ell_1 > \ell_2 > \dots > \ell_p > 0$ and $D_\lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_p > \lambda_{p-1} > \dots > \lambda_1 > 0$ respectively. If

$$c_{ij} = (\lambda_j - \lambda_i)(\ell_i - \ell_j) / (1 + \lambda_i \ell_i)(1 + \lambda_j \ell_j) \text{ for } i \neq j, i, j = 1, 2, \dots, p,$$

then the asymptotic joint distribution of $\ell_1, \ell_2, \dots, \ell_p$ is

$$f_0(D_\ell | D_\lambda) [1 + \frac{1}{2n} (\sum_{i < j} c_{ij}^{-1} + \frac{1}{12} p(p-1)(4p+1)) + O(n^{-2})]$$

where $f_0(D_\ell | D_\lambda)$ is defined by (4.9).

4.2 Last $(p-q)$ Population Roots Equal

Now, let us consider a situation when $D_\lambda = \text{diag}(\Lambda, \lambda I_{p-q})$, and

$$\lambda > \lambda_q > \dots > \lambda_1 > 0.$$

Let $H = \begin{pmatrix} H_1 \\ H_2 \end{pmatrix}^q_{p-q}$ in (4.1). Then

$$|I_p + D_\lambda H D_\ell H'| = |I_p + \lambda D_\ell + H'_1(\Lambda - \lambda I_q)H_1 D_\ell|.$$

Transform Q to H and use Corollary 2.1, equation (2.4), for integration over H_2 for given H_1 . Then, use Corollary 2.2 to transform H_1 to Q_{11} and Q_{12} as in Section 3. With this operation, we obtain from (4.1) the joint density of $\ell_1, \ell_2, \dots, \ell_p$ as

$$(4.18) \quad \frac{\pi^{\frac{1}{2}(p-q)^2} |D_\lambda|^{\frac{1}{2}n_1} |D_\ell|^{\frac{1}{2}(n_1-p-1)} \alpha_p(D_\ell)}{\Gamma_{p-q}(\frac{p-q}{2}) n^{\frac{1}{2}pq - \frac{1}{4}q(q-1)} \beta_p(\frac{1}{2}n_1, \frac{1}{2}n_2)} \\ \cdot \int_{T_{11}} \int_{T_{12}} |I_p + D_\lambda H_0 D_\ell H'_0|^{-\frac{1}{2}n} \\ \cdot |I - (4n)^{-\frac{1}{2}} T|^{-(p-1)} dT_{11} dT_{12}$$

with $T = \begin{pmatrix} T_{11} & T_{12} \\ -T_{12} & 0 \end{pmatrix}$ a skew-symmetric matrix and $H_0 = -I + 2(I - (4n)^{-\frac{1}{2}} T)^{-1}$.

Using the method given above, we get the following results after making use of the following modifications in the development:

In the expression (4.13), we substitute $T_{22} = 0$, and hence write

$$c_{ij} = (\lambda_j - \lambda_i)(\ell_i - \ell_j)/(1 + \lambda_i \ell_i)(1 + \lambda_j \ell_j) \quad \text{for } i \neq j, i, j = 1, 2, \dots, q \\ = (\lambda - \lambda_i)(\ell_i - \ell_j)/(1 + \lambda \ell_j)(1 + \lambda_i \ell_i) \quad \text{for } j = q+1, \dots, p \text{ and } i = 1, 2, \dots, q.$$

Define

$$(4.19) \quad f_{01}(D_\ell | D_\lambda) = \\ \frac{(2\pi/n)^{\frac{1}{2}pq - \frac{1}{4}q(q+1)} \pi^{\frac{1}{2}(p-q)^2} |D_\ell|^{\frac{1}{2}(n_1-p-1)} |I + D_\lambda D_\ell|^{-\frac{1}{2}(n-p+1)} (\alpha_p(D_\ell))^{\frac{1}{2}}}{\beta_p(\frac{1}{2}n_1, \frac{1}{2}n_2) \Gamma_{p-q}(\frac{p-q}{2}) \left[\prod_{i=1}^q (\lambda - \lambda_i) \right]^{\frac{1}{2}(p-q)} \left[\prod_{j=q+1}^p (1 + \lambda \ell_j) \right]^{\frac{1}{2}(p-q-1)} \left[\prod_{j=i+1}^q (\lambda_j - \lambda_i) \right]^{\frac{1}{2}}}$$

and the density of T_{11}, T_{12} as given in (3.23). Then (4.18) can be written as

$$(4.20) \quad f_{01}(D_\ell | D_\lambda) \left[1 + \sum_{j=1}^{\infty} n^{-j} a_j \right]$$

where $a_j = E g_{2j}(T_{11}, T_{12})$ where $g_{2j}(T_{11}, T_{12})$ is the same as defined in (3.15) with the same notations of section 4 for $\{T\}_j$ and $T_{22} = 0$. To obtain a_1 explicitly, we have to put $t_{ij} = 0$ for $ij = q+1, \dots, p$ ($i \neq j$), in the development of $\{T\}_3$ and $\{T\}_4$ given in (4.15) and (4.16), and we get

$$4E \{T\}_3^2 = \left(\sum_{k=q+1}^p \sum_{i < j} + \sum_{i < j < k=1}^q \right) \{ (c_{ij}c_{ik}c_{jk})^{-1} \\ [c_{ij}^2 + c_{ik}^2 + c_{jk}^2 - 2(c_{ij}c_{ik} + c_{ij}c_{jk} + c_{ik}c_{jk}) - 4c_{ij}c_{ik}c_{jk}] \}, \\ E(\text{tr } T^2) = -2 \sum_{i < j}^q (c_{ij}^{-1}) - 2 \sum_{j=q+1}^p \sum_{i=1}^q c_{ij}^{-1}$$

and

$$8E \{T\}_4 = 6 \left(\sum_{i < j=1}^q c_{ij}^{-2} (c_{ij} + c_{ij}^2) + \sum_{j=q+1}^p \sum_{i=1}^q c_{ij}^{-1} (c_{ij} + c_{ij}^2) \right) \\ + \left(\sum_{k=q+1}^p \sum_{i < j=1}^q + \sum_{i < j < k=1}^q \right) \left\{ \frac{1}{c_{ij}c_{jk}} (2c_{ij} + 2c_{jk} - c_{ik} + 4c_{ij}c_{jk}) \right. \\ + (c_{ij}c_{ik})^{-1} (2c_{ij} + 2c_{ik} - c_{jk} + 4c_{ij}c_{ik}) \\ \left. + (c_{ik}c_{jk})^{-1} (2c_{ik} + 2c_{jk} - c_{ij} + 4c_{ik}c_{jk}) \right\} \\ + \sum_{j < k=q+1}^p \sum_{i=1}^q \frac{1}{c_{ij}c_{ik}} [2c_{ij} + 2c_{ik} - c_{jk} + 4c_{ij}c_{ik}].$$

This expression leads to

$$a_1 = E g_2(T_{11}, T_{12}) = E \{T\}_4 + \frac{1}{2} E \{T\}_3^2 + \frac{p-1}{8} E(\text{tr } T^2) \\ = \frac{1}{2} \left(\sum_{i < j=1}^q (c_{ij})^{-1} + \sum_{j=q+1}^p \sum_{i=1}^q c_{ij}^{-1} \right) \\ + \frac{1}{24} q(q-1)(4q+1) + \frac{1}{4} q(p^2 - q^2).$$

Hence, we get the following

Theorem 4.2

In Theorem 4.1, let $\lambda_{q+1} = \dots = \lambda_p = \lambda$ (say),

$c_{ij} = (\lambda_j - \lambda_i)(\ell_i - \ell_j)/(1 + \lambda_i \ell_i)(1 + \lambda_j \ell_j)$ for $i \neq j = 1, 2, \dots, q$ and

$c_{ij} = (\lambda - \lambda_j)(\ell_i - \ell_j) / (1 + \lambda \ell_j)(1 + \lambda_i \ell_i)$ for $i = 1, 2, \dots, q$ and $j = q+1, \dots, p$.

Then, the joint density of $\ell_1, \ell_2, \dots, \ell_p$ for large $n = n_1 + n_2$ is given by

$$f_{01}(D_\ell | D_\lambda) \left[1 + \frac{1}{2n} \left(\sum_{i < j=1}^q c_{ij}^{-1} + \sum_{j=q+1}^p \sum_{i=1}^q c_{ij}^{-1} + \frac{1}{12} q[(q-1)(4q+1) + 6(p^2 - q^2)] + O(n^{-2}) \right] \right]$$

where

where $f_{01}(D_\ell | D_\lambda)$ is defined in (4.19).

Theorem 4.2 can be generalized to the situation when

$D = (\delta_1 I_{p_1}, \dots, \delta_k I_{p_k}), p_1 + \dots + p_k = p$ and at least one of $p_i (i = 1, 2, \dots, k)$ is unity.

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